Global Mortality Trends (1990-2019): A Pre-Pandemic Baseline

Student Name:

Student ID:

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Abstract

This report presents a visual analysis of the "Global Cause of Deaths" dataset (1990-2019), aiming to establish a definitive pre-pandemic baseline for global health. Applying the What-Why-How visualization framework, the study elucidates the complex mortality patterns that defined the world immediately prior to the COVID-19 crisis. Core objectives include identifying longitudinal trends, mapping geospatial distributions, and utilizing machine learning to classify national health profiles. Through time-series charts, choropleth maps, and K-Means clustering, the analysis captures the completion of the epidemiological transition in developed nations and the persistent "double burden" in developing regions. These findings provide a critical historical reference point for understanding the systemic shifts caused by the subsequent global pandemic.

1. Introduction

1.1 Research Background

Global mortality patterns reflect public health and social development. The period from 1990 to 2019 saw rapid globalization and major health transitions, ending just before COVID-19. Studying this era provides a baseline to assess post-2020 health changes. Using the Global Burden of Disease framework, this project analyzes pre-pandemic mortality trends and the socioeconomic factors influencing them.

1.2 Research Objectives

The main aim of this project is to use data visualization and machine learning to develop a comprehensive overview of global mortality before 2020. The analysis addresses four questions:

- **Global Epidemiological Transition:** How did the balance between non-communicable and CMNN diseases evolve to form the 2019 health structure?
- **Socioeconomic Inequalities:** How did mortality burdens differ across World Bank income groups?
- **Disease Relationships:** What are the statistical associations between different causes of death?
- **Health Clusters:** Can countries be grouped into distinctive clusters based on mortality profiles rather than economic indicators?

2. Data Description and Methodology

2.1 Data Description

This study combines four datasets to examine the relationship between mortality, demographics, and development. The main dataset, Cause of Deaths around the World, obtained from Kaggle, contains annual mortality records from 1990 to 2019 for twenty-nine causes. To support normalization and allow socioeconomic comparisons, three additional indicators were collected from the World Bank. GDP per capita reflects the level of economic development. Total population is used for standardization across countries. The World Bank Income Group Classification identifies each country as Low, Lower-Middle, Upper-Middle, or High Income.

2.2 Methodology

All data processing, calculations, and visualizations were conducted using R (for longitudinal analysis) and Python (for machine learning clustering). The workflow comprised four stages.

2.2.1 Integration

The dataset was expanded by merging World Bank income group information and GDP data for each country, allowing stratified comparisons across economic levels.

2.2.2 Standardization

To correct for population bias and enable fair international comparisons, raw mortality counts were converted into two key indicators:

- Mortality Rate per 100,000 people: Measures the risk intensity, supporting direct comparison across countries and years.
- Proportion of Total Deaths: Reflects the relative structure of mortality within each region.

2.2.3 Classification

Causes of death were grouped following the Global Burden of Disease (GBD) framework:

- Non-Communicable Diseases (NCDs): Chronic conditions such as cardiovascular diseases and cancers.
- CMNN: Communicable, Maternal, Neonatal, and Nutritional diseases.
- Injuries: External causes including road injuries, violence, and conflict.

2.2.4 Analytical Approach

The analysis followed a hierarchical structure from descriptive to machine learning methods:

- Longitudinal Trends: Examining the global epidemiological transition from 1990 to 2019.
- Socioeconomic Stratification: Comparing mortality profiles across World Bank income groups to reveal disparities.
- Correlation & Variance: Using Spearman correlation matrices and boxplots to explore disease clustering and global inequality.
- Machine Learning Clustering: K-Means clustering was applied to 2019 data to define a pre-pandemic baseline, categorizing countries into distinct health clusters based solely on their 29-cause mortality profiles, independent of economic labels.

3. Results and Analysis

3.1 The Global Epidemiological Transition

This section describes the epidemiological transition, which refers to the global shift from infectious diseases to chronic diseases during the past thirty years.

3.1.1 The Structural Shift

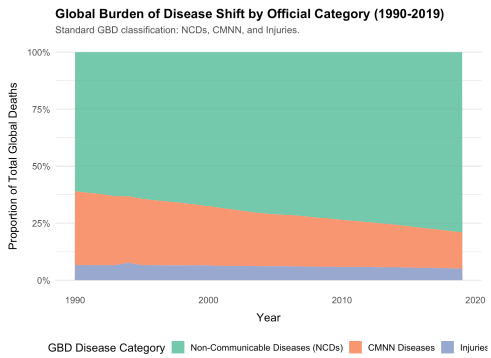


Figure 3.1

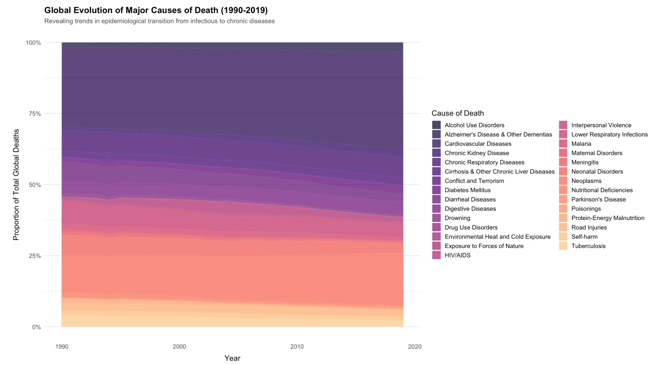


Figure 3.2

Figure 3.1 shows a clear change in the global mortality structure. In 1990, Communicable, Maternal, Neonatal, and Nutritional (CMNN) diseases made up about 33% of all deaths. By 2019, this share had fallen to below 20%. During the same period, Non-Communicable Diseases (NCDs) increased from 58% to 74%, becoming the main drivers of global mortality.

Figure 3.2 offers a more detailed view of this process. It illustrates how infectious causes (shown in warm colors) gradually decreased, while chronic conditions such as Neoplasms and Cardiovascular Diseases (shown in cool colors) continued to expand and dominate the overall pattern.

3.1.2 The Changing Hierarchy of Death

To measure the specific changes within these broad categories, Figure 3.3 presents the change in ranking for the top 15 causes of death from 1990 to 2019.

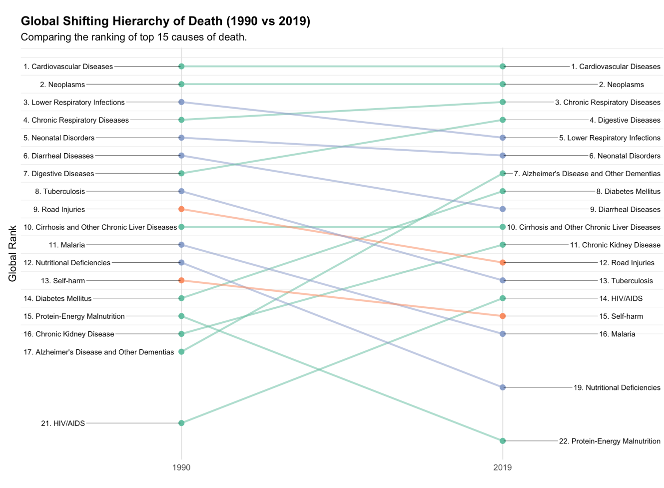


Figure 3.3

The crossing lines highlight a major shift in the global mortality landscape.

CMNN causes (blue lines) show clear declines. For example, Diarrheal Diseases moved from Rank 4 in 1990 to Rank 9 in 2019, reflecting improvements brought by global public health interventions.

In contrast, NCDs (green lines) generally moved upward. Alzheimer's Disease recorded the most notable rise, increasing from Rank 10 to Rank 5.

At the highest level, cardiovascular diseases and Neoplasms remained in Rank 1 and Rank 2 respectively, demonstrating their long-term and stable impact on global mortality.

3.2 Geographic and Socioeconomic Disparities

Global aggregate trends often conceal the substantial inequalities that persist across regions. This section breaks down the global average to show how the epidemiological transition differs by geography and by level of economic development.

3.2.1 Geospatial Distribution: Global Regional Differences

The choropleth maps (Figures 3.4-3.6) illustrate clear regional differences in the global burden of disease.

Non-Communicable Diseases dominate the mortality structure in higher-income regions such as North America, Europe, and Australia. These areas show a strong concentration of NCDs, which reflects aging populations and lifestyle-related risks.

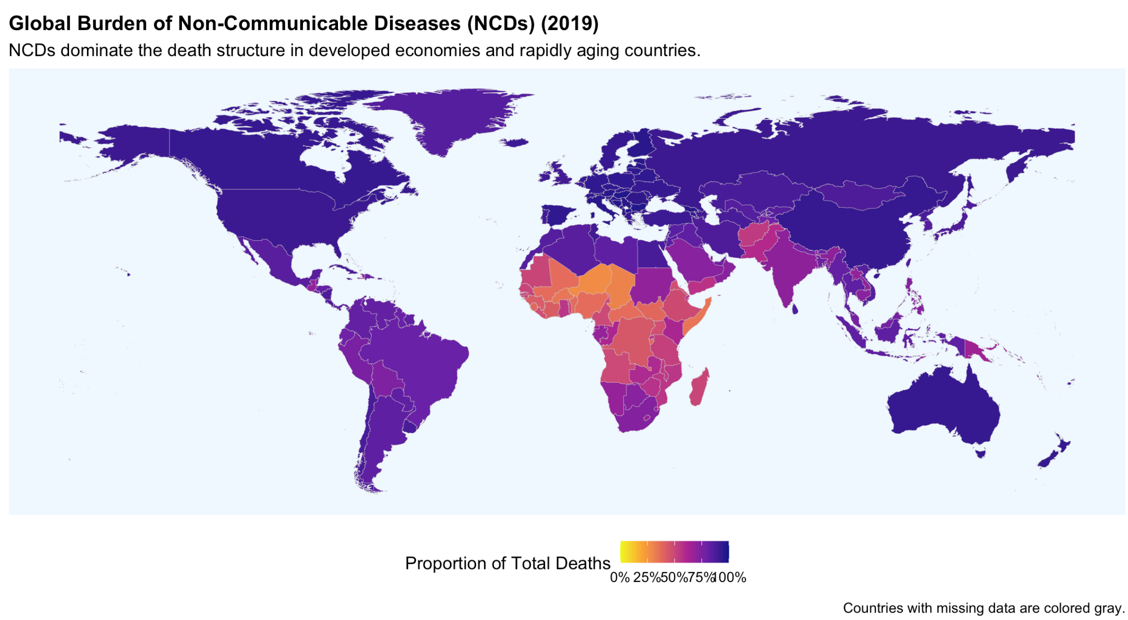


Figure 3.4

By contrast, Communicable, Maternal, Neonatal, and Nutritional (CMNN) diseases remain heavily concentrated in Sub-Saharan Africa and parts of South Asia. This distribution shows that for many lower-income regions, preventable infectious diseases continue to represent the primary threat to health.

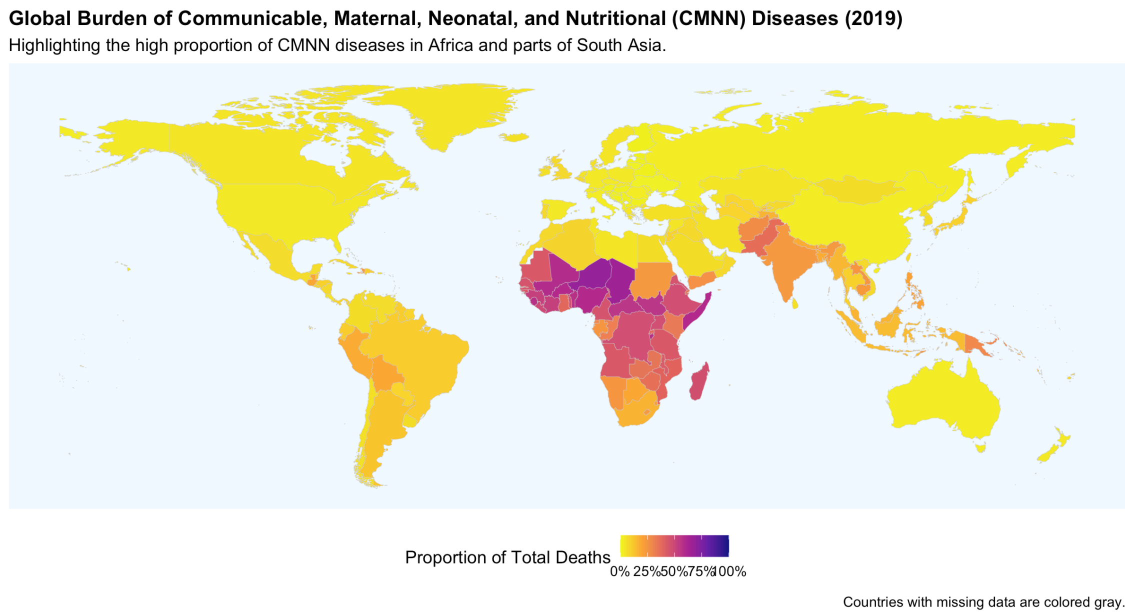


Figure 3.5

The pattern of Injuries is more heterogeneous, with higher burdens in conflict-affected regions and in areas with developing infrastructure, where external hazards play a greater role than biological illness.

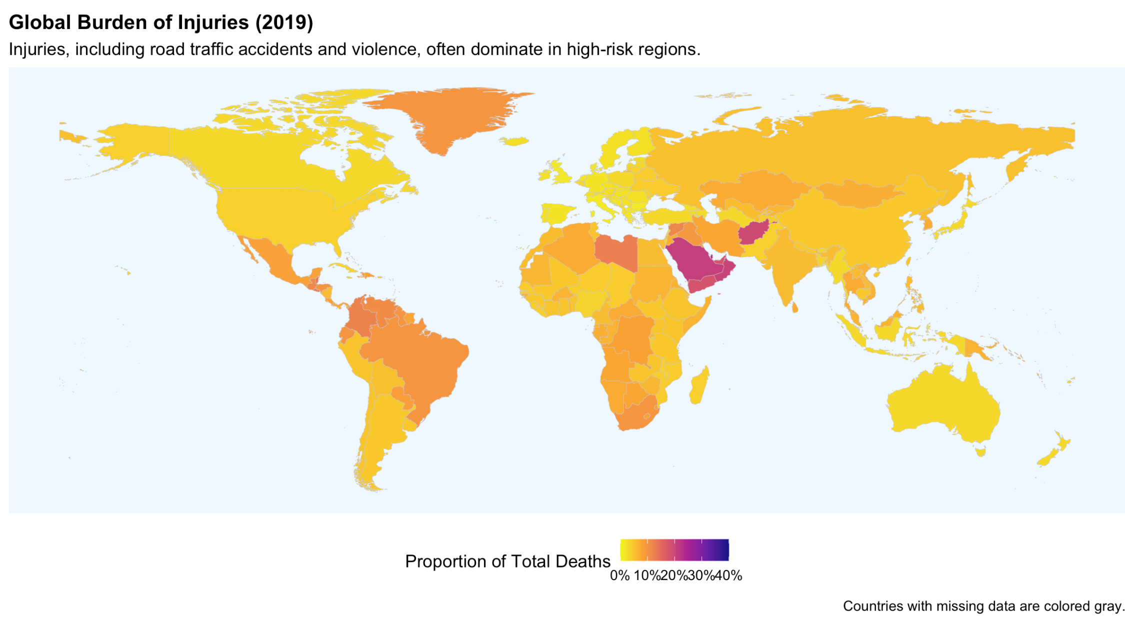
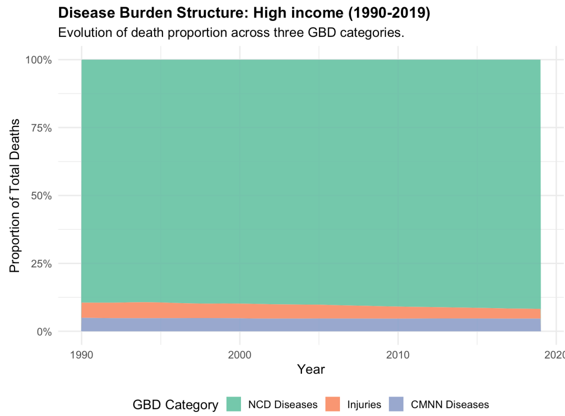


Figure 3.6

3.2.2 Socioeconomic Stratification: Evolutionary Differences by Income

Comparing the high-income and low-income groups reveals two distinct epidemiological paths.



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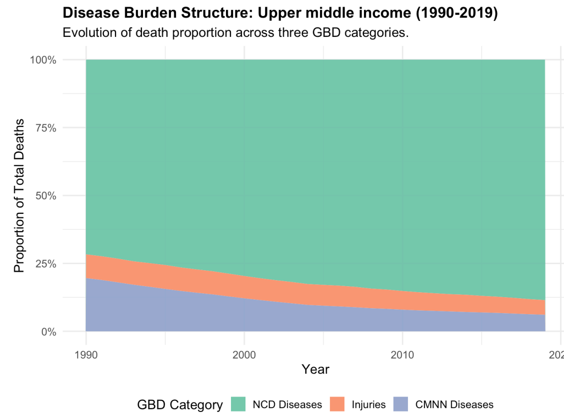

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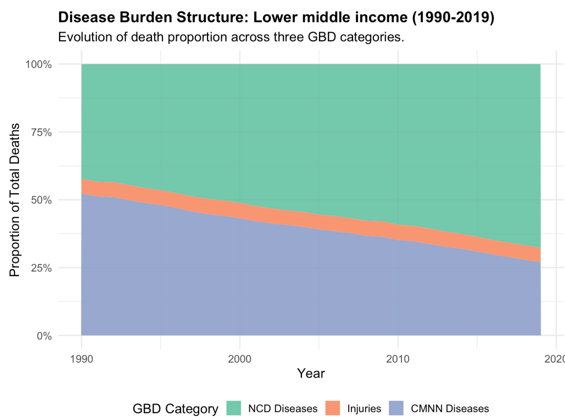

Figure 3.10

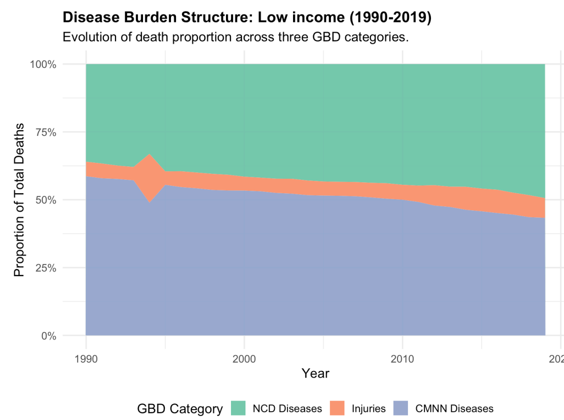

Figure 3.11

High-income nations (Figure 3.7) reached a post-transitional stage earlier, and their mortality structure has remained relatively stable since 1990. Non-Communicable Diseases consistently account for more than 85 percent of all deaths, forming a predictable and aging-related pattern.

Low-income nations (Figure 3.10) present a very different profile. Their disease structure has changed more rapidly, and although the share of CMNN diseases has declined, these causes remained dominant well into the 2000s. These countries face a double burden of persistent infectious diseases and rising chronic conditions.

A notable feature in Figure 3.10 is the sharp spike in the Injuries category around 1994. This anomaly corresponds to the Rwandan Genocide and demonstrates how geopolitical instability can sharply disrupt health outcomes in low-income settings, a level of volatility rarely observed in wealthier economies.

3.3 Macro-Factor Correlation and Global Variance

This section examines how broader structural factors shape global disease patterns through correlation analysis and cross-country variance.

3.3.1 Disease Clustering: Syndemic Patterns

A correlation matrix was calculated for all 29 causes. The heatmap shows two clear clusters.

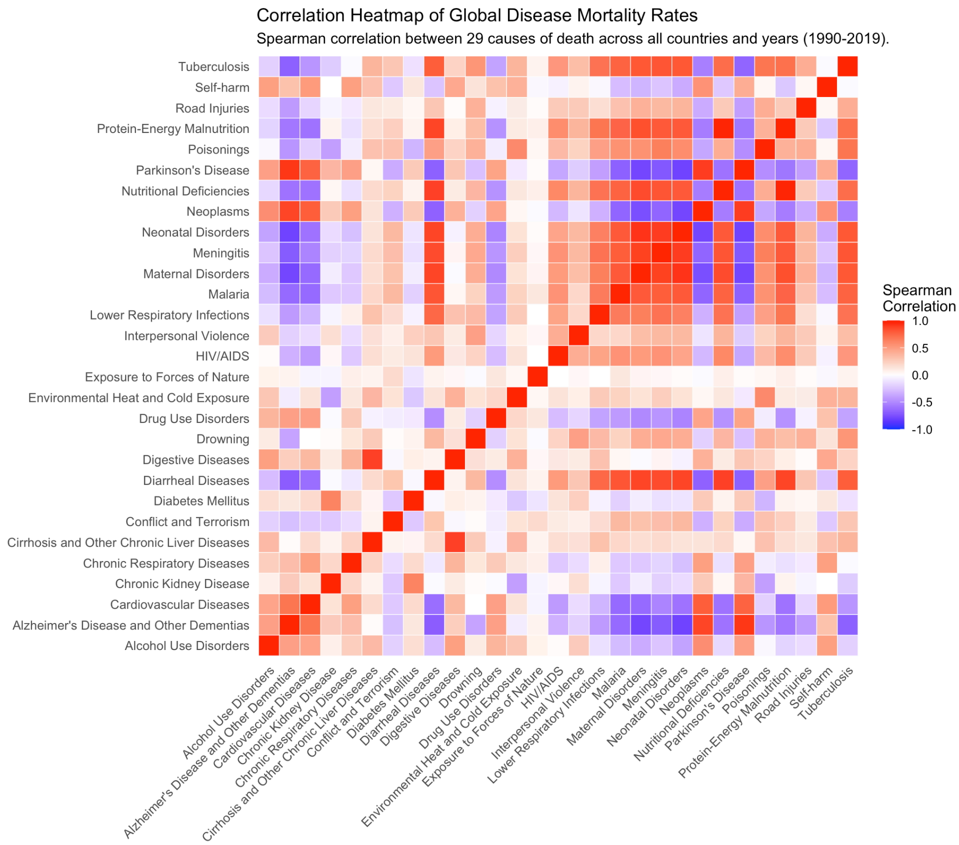

Figure 3.11

The first cluster consists of infectious causes such as Malaria, Diarrheal Diseases, and Meningitis. These causes strongly correlate with one another, suggesting that countries facing one infectious threat often face several at the same time due to shared infrastructure limitations.

A second cluster appears among chronic conditions, including Alzheimer's Disease and Colorectal Cancer, which co-occur more often in higher-income settings. The two clusters show strong negative correlations. Countries with high infectious burdens tend to have low levels of degenerative NCDs, while countries with advanced control of infections show higher rates of age-related chronic diseases. This contrast reflects the progression of the epidemiological transition.

3.3.2 Global Variance and Inequality

To assess global inequality, mortality rate distributions for all countries in 2019 were examined. The CMNN group displays the greatest variance. Several infectious causes show long boxes and wide whiskers, indicating that some countries have nearly eliminated these diseases while others still face very high mortality levels.

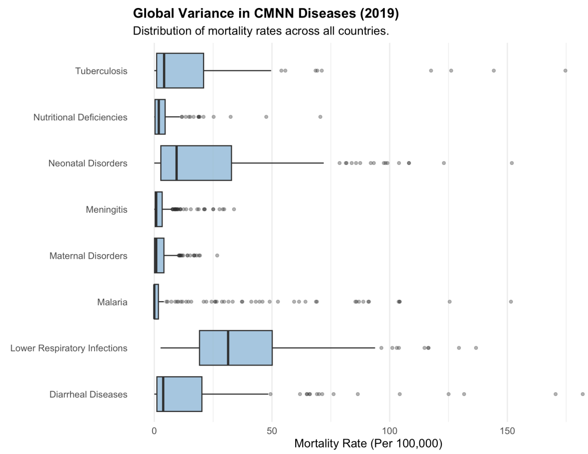

Figure 3.12

NCDs have higher median rates but much tighter spreads. Cardiovascular Diseases and Neoplasms affect all regions and show far less disparity.

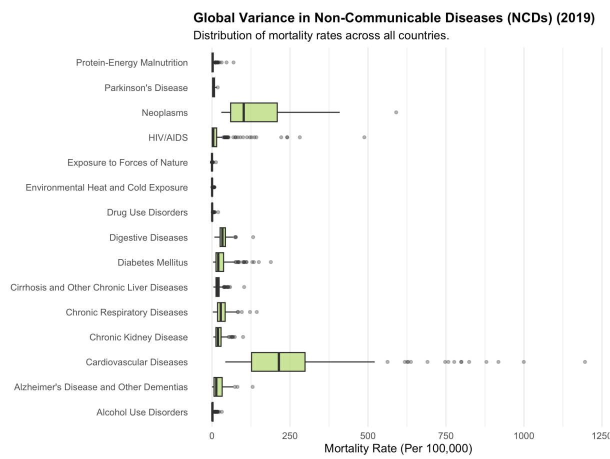

Figure 3.13

Injuries present moderate variance. Road Injuries in particular show a wide distribution, reflecting uneven levels of infrastructure safety and traffic regulation worldwide.

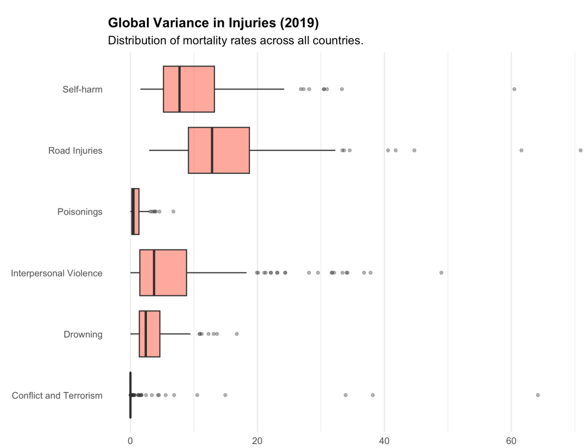

Figure 3.14

3.4 Case Study: The Rapid Health Transition in China

China offers a clear example of how economic growth can reshape a nation's mortality structure within a short period.

3.4.1 The Compressed Transition

China completed in three decades a shift that took many Western countries more than a century. Figures 3.15 - 3.16 show both the structural change and the reordering of major causes of death.

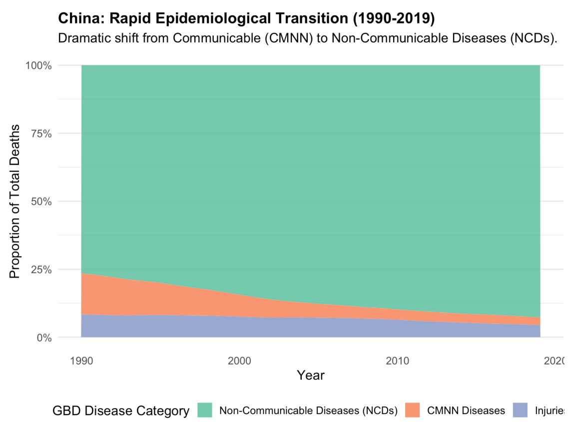

Figure 3.15

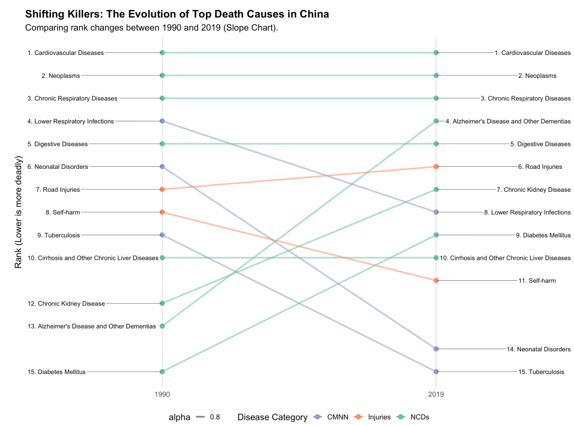

Figure 3.16

Figure 3.15 illustrates the decline of infectious and maternal causes. In 1990, these diseases still formed a visible share of deaths. By 2019, they had been largely replaced by Non-Communicable Diseases (NCDs), which now dominate the mortality structure.

Figure 3.16 tracks changes in specific rankings. Lower Respiratory Infections, once the leading cause of death, fell sharply in rank due to improvements in sanitation and healthcare. Meanwhile, Stroke and Ischemic Heart Disease remained top killers, and Lung Cancer moved upward, reflecting patterns typical of developed economies.

3.4.2 Economic Drivers and Global Positioning

Figures 3.17 - 3.18 explore the factors behind this shift and show China's place in the global landscape.

Figure 3.17 plots GDP per capita against the mortality rate for cardiovascular diseases. The two curves rise in parallel, indicating that rapid economic growth reduced infectious risks but introduced lifestyle-related health challenges.

Figure 3.18 places China on a global scatter plot. Although China was still a middle-income economy in 2019, its cardiovascular mortality level aligned more with high-income countries. This suggests that China now carries a "developed" disease burden before reaching developed-country income levels, creating significant pressure for future health planning.

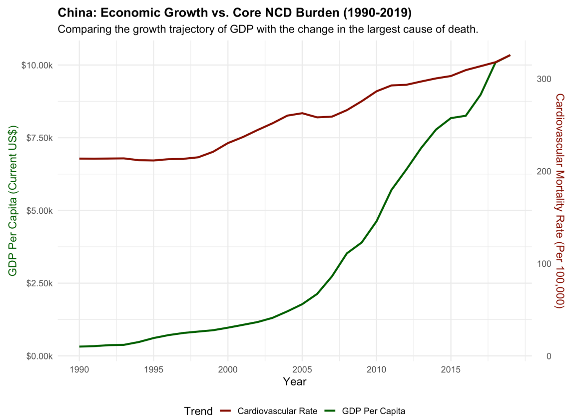

Figure 3.17

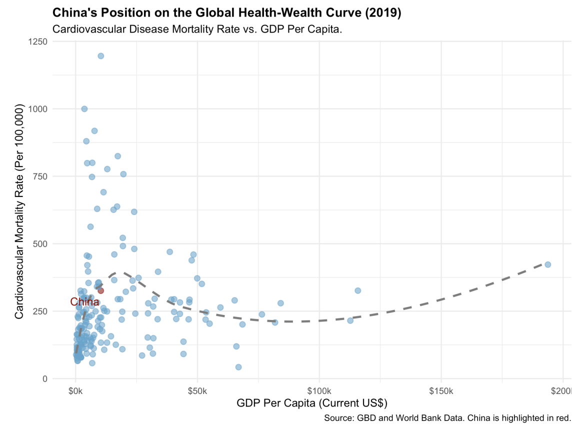

Figure 3.18

3.5 Machine Learning Analysis: The Pre-Pandemic Baseline (2019)

This section shifts from long-term trends to a cross-sectional view of the year 2019. Using K-Means Clustering, an unsupervised machine learning method, we group countries based on their mortality patterns in the final year of the dataset. The goal is to build a structural baseline of global health just before COVID-19.

3.5.1 Unsupervised Classification: Identifying Three Global Health Patterns

The model was trained using mortality rates from all 29 causes as input features. It was instructed to form three clusters without using income or geographic labels. The resulting world map shows a clear three-way division of countries according to their disease burdens.

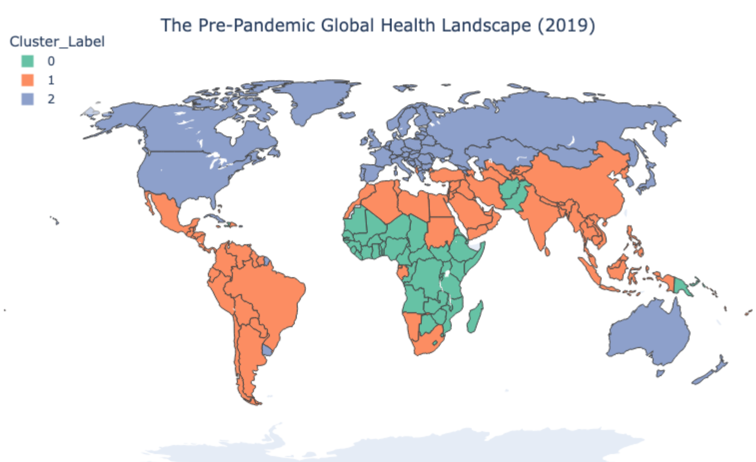

Figure 3.19

3.5.2 Cluster Profiles: Distinct Health Patterns

To understand these clusters, we examined the feature centroids with a radar chart. Each cluster displays a unique mortality profile.

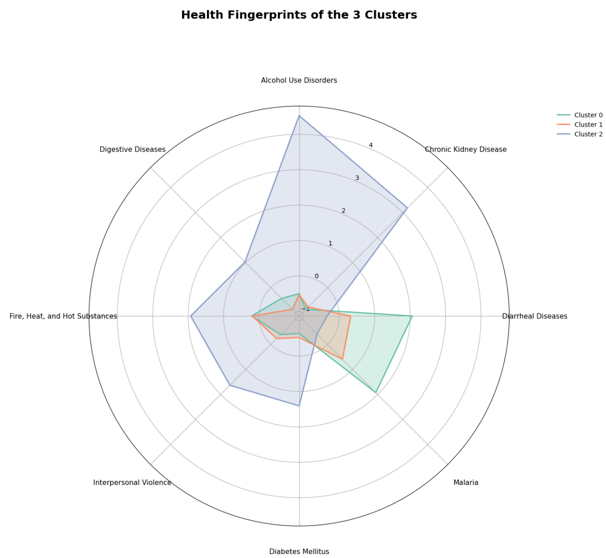

Figure 3.20

- Cluster 0 ("Infectious" Cluster - Green): Concentrated in Sub-Saharan Africa and parts of South Asia. This group shows extremely high mortality from preventable infectious diseases such as Malaria and Diarrheal Diseases. It represents the world's remaining infectious-disease burden.
- Cluster 2 ("Affluent/Aging" Cluster - Purple): Covering North America, Western Europe, Australia, and Japan. These countries have very low infectious mortality but high levels of degenerative diseases, including Alzheimer's Disease and Colorectal Cancer. The pattern reflects aging and long-life expectancy.
- Cluster 1 ("Transitional/Mixed" Cluster - Orange): Including Latin America, Russia, and several Middle Eastern countries. These nations carry a double burden: lingering infectious diseases combined with rising chronic conditions. They also show higher mortality from external causes such as Interpersonal Violence.

China sits on the boundary between the purple and orange clusters. Its position shows that by 2019; China had moved far away from the infectious-disease profile of low-income countries. Statistically, it aligns more with affluent or transitional health patterns, supporting the findings from the previous case study.

4. Discussion and Conclusion

4.1 Key Findings

This visual analysis of global mortality (1990-2019) highlights a major epidemiological shift through four perspectives:

The Structural Crossover (Global Trends): Stacked area charts confirm a decisive transition worldwide. Non-Communicable Diseases (NCDs) have overtaken CMNN diseases as the leading causes of death. Slope charts further validate this trend, showing sharp declines in infectious killers such as Diarrheal Diseases, while age-related conditions like Alzheimer's Disease have risen to the top ranks.

The "Double Burden" of Inequality (Disparities): Geographic and variance analyses reveal that progress is uneven. NCDs have become a global challenge, but Cluster 0 in the K-Means analysis highlights regions still trapped in a syndemic of preventable infectious diseases. Sub-Saharan Africa remains statistically isolated from global improvements.

The Compressed Transition (Case Study): China demonstrates a rapid epidemiological shift. Unlike the century-long transition in Western nations, China compressed this evolution into three decades. The close correlation between GDP growth and cardiovascular mortality suggests that rapid economic development reduces infectious risks but introduces lifestyle-related health burdens.

Health Clusters Beyond Geography (Machine Learning): Unsupervised learning shows that national health profiles are not strictly determined by location. The three clusters identified—Infectious, Transitional, and Affluent—provide a data-driven framework to categorize countries based on disease burden rather than economic status alone.

4.2 Limitations

Data Estimation: The analysis relies on GBD estimates, which may be affected by under-reporting, especially in countries with weaker vital registration systems.

Temporal Scope: Ending in 2019 establishes a pre-pandemic baseline, but the analysis does not account for COVID-19, which likely caused temporary reversals in communicable disease trends.

4.3 Conclusion

This report establishes a clear pre-pandemic baseline for global health. By 2019, historical infectious diseases had largely declined, while chronic and degenerative conditions were increasingly dominant. This structural snapshot provides a critical reference for future research to assess the impact of COVID-19, not only in absolute mortality but also in terms of deviations from long-term structural trends.

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